

Student Enrollment & Course Management System (SECMS)

User Story:

The Student Enrollment & Course Management System is a comprehensive Salesforce application designed to streamline and automate the academic enrollment operations of an educational institute. This system enables administrators, enrollment officers, and instructors to efficiently manage student records, course offerings, enrollments, instructor assignments, approval processes, payments, and reporting—all within a unified Salesforce environment.

1. INTRODUCTION

1.1 Project Overview :

This project aims to centralize all student, course, and enrollment-related data within a single Salesforce application to ensure easy data access, structured workflows, and accurate reporting. The system allows Enrollment Officers to manage student admissions, course registrations, and communication effectively, while instructors and administrators gain full transparency into course allocations and academic progress. The solution also enables automated instructor assignment, task creation, validation checks, and multi-level approval processes for enrollment management.

With Salesforce Flows, Approval Processes, Validation Rules, Reports, and Dashboards, the organization can ensure that all enrollment operations—from admission to course completion—are handled systematically and with continuous monitoring of key performance indicators such as enrollment volume, revenue, student status distribution, and pending approvals.

1.2 Purpose :

The primary purpose of SECMS is to replace manual and fragmented enrollment processes with a scalable, automated, and secure Salesforce solution that ensures:

Accurate student data management

Structured enrollment workflows
Automated approvals and validations
Transparent reporting and monitoring

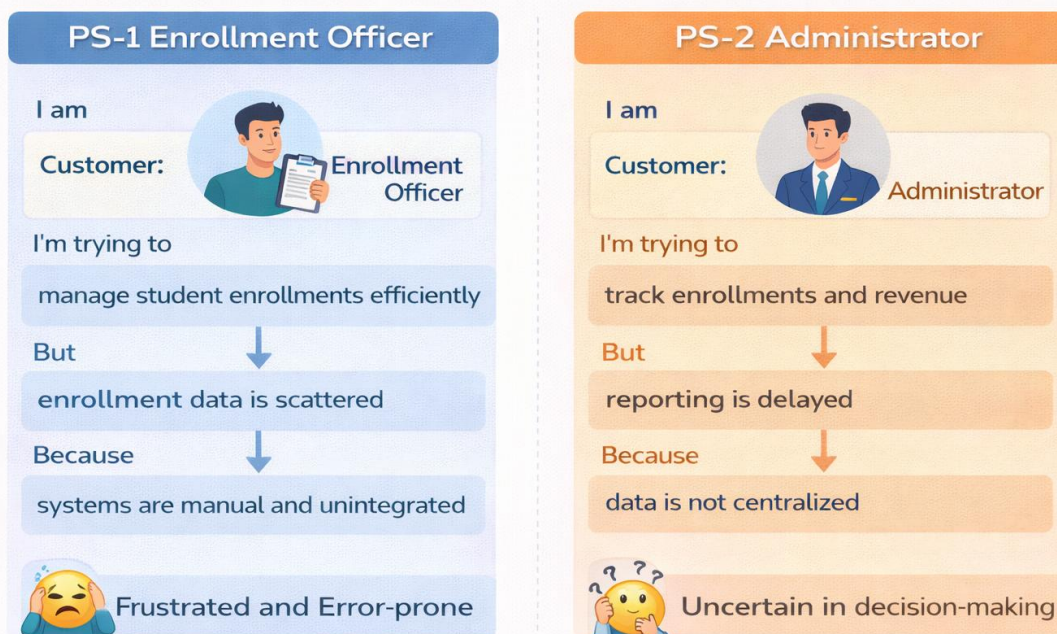
2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Enrollment Officer	Manage student enrollments efficiently	enrollment data is scattered	systems are manual and unintegrated	frustrated and error-prone
PS-2	Administrator	track enrollments and revenue	reporting is delayed	data is not centralized	uncertain in decision- making

2. IDEATION PHASE

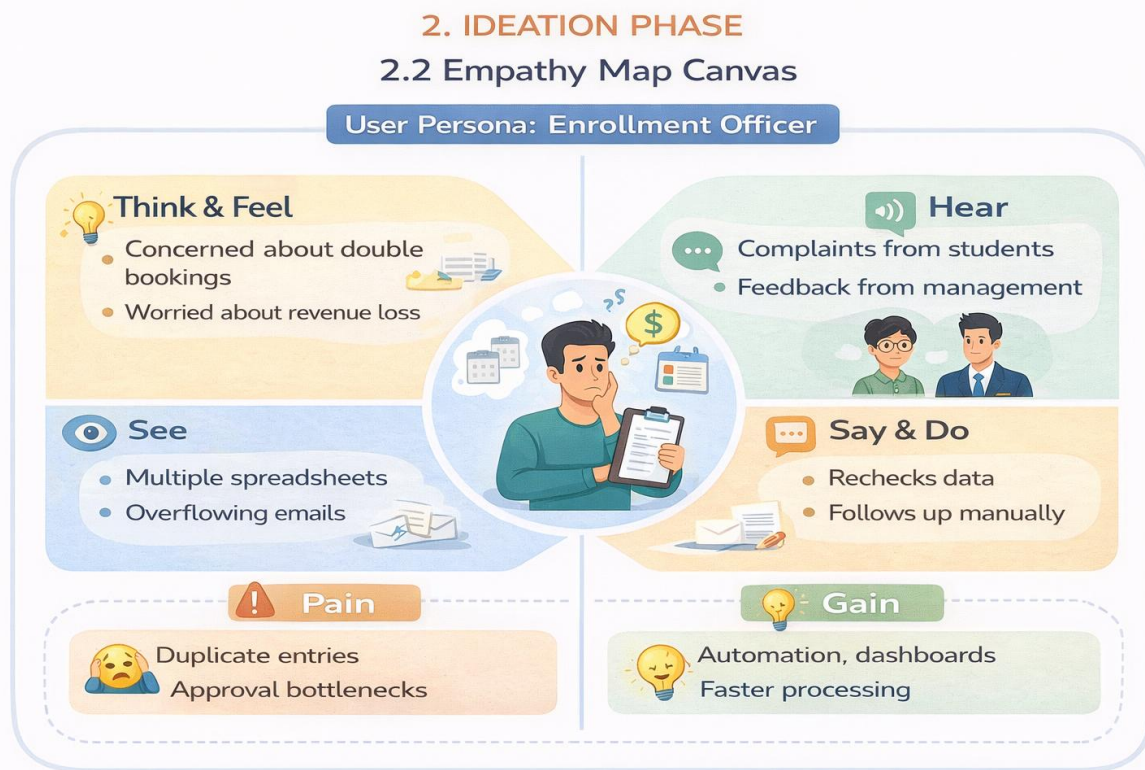
2.1 Problem Statement (Customer Problem Statement Template)



2.2 Empathy Map Canvas

User Persona: Enrollment Officer

- **Think & Feel:** Concerned about double bookings and revenue loss
- **Hear:** Complaints from students and management
- **See:** Multiple spreadsheets and emails
- **Say & Do:** Rechecks data, follows up manually
- **Pain:** Duplicate entries, approval bottlenecks
- **Gain:** Automation, dashboards, faster processing



2.3 Brainstorming & Idea Prioritization

Step 1 – Brainstormed Ideas

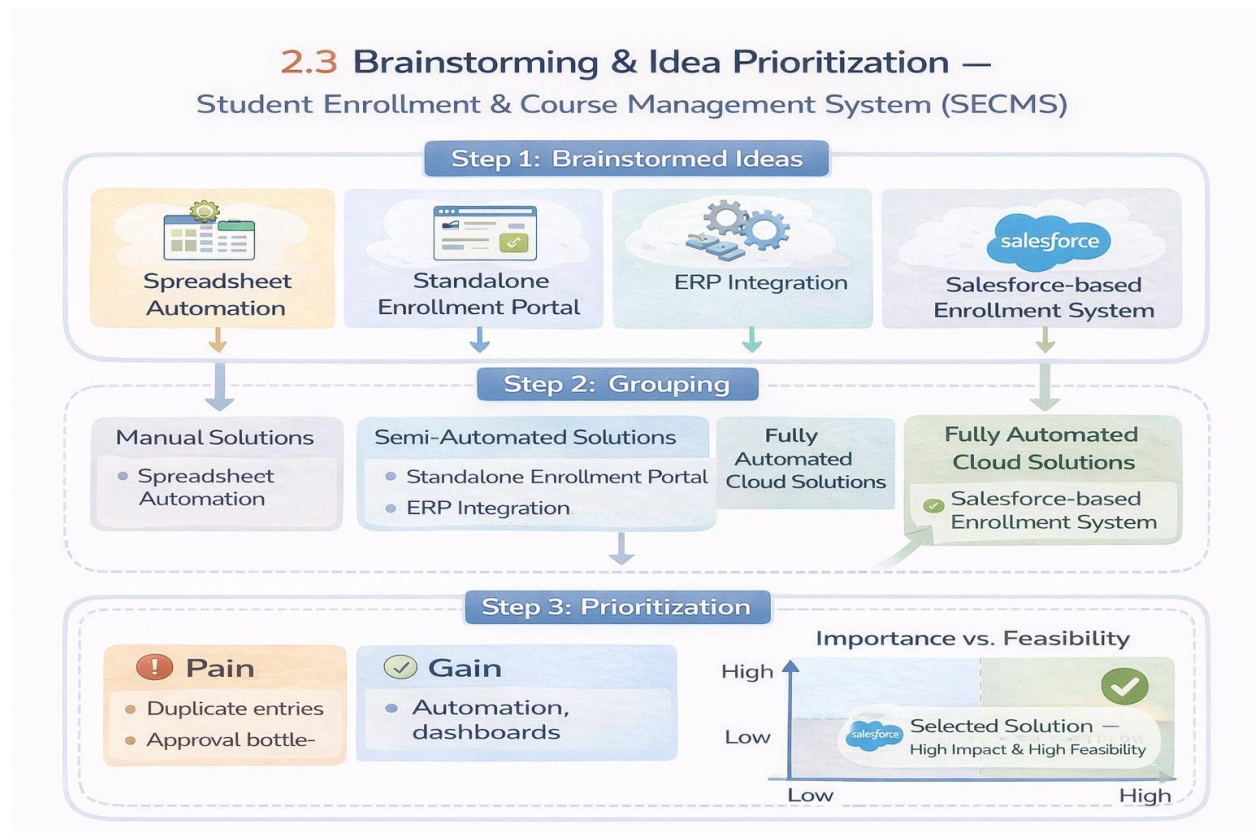
- Spreadsheet automation
- Standalone enrollment portal

- ERP integration
- Salesforce-based enrollment system

Step 2 – Grouping

- Manual solutions
- Semi-automated solutions
- Fully automated cloud solutions

Step 3 – Prioritization
Salesforce-based SECMS was selected due to **high feasibility** and **high impact**.



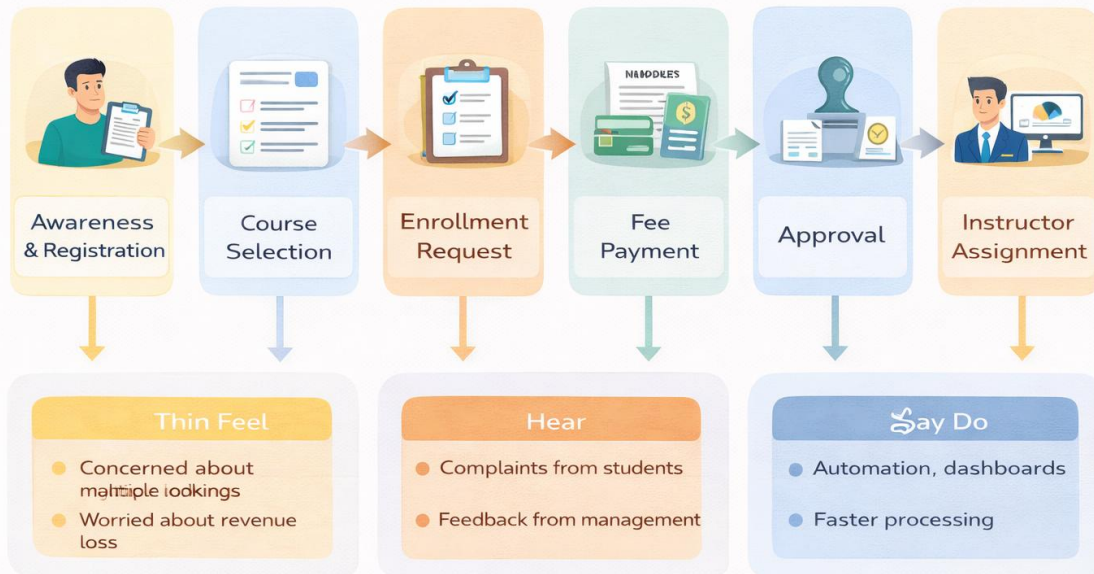
3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages covered:

- Awareness & Registration
- Course Selection
- Enrollment Request
- Fee Payment
- Approval
- Instructor Assignment
- Reporting & Completion

3.1 Customer Journey Map – Student Enrollment & Course Management System (SECMS)



3.2 Solution Requirements

Functional Requirements

FR No	Epic	Description
FR-1	Student Management	Manage student records

FR-2	Course Management	Create and manage courses
FR-3	Enrollment Processing	Request, approve, reject enrollments
FR-4	Instructor Assignment	Auto-assign instructors
FR-5	Reporting	Dashboards and reports

Non-Functional Requirements

NFR-1: Usability

The system shall provide a simple, intuitive, and user-friendly interface that enables enrollment officers, administrators, instructors, and students to perform operations efficiently with minimal training. Navigation shall be organized through clearly labeled tabs, dashboards, and forms to reduce user effort and errors.

Examples:

- Quick student registration form
- One-click enrollment approval
- Role-based dashboards

NFR-2: Security

The system shall ensure the confidentiality, integrity, and protection of student and institutional data through secure authentication, authorization, and access controls. Sensitive information such as personal data, enrollment history, and academic records shall only be accessible to authorized users.

Examples:

- Only admins can modify course details
- Instructors view only assigned students
- Students cannot access other student records

NFR-3: Reliability

The system shall operate consistently and accurately without failures during enrollment processing, approvals, and reporting. Automation logic shall execute correctly to avoid duplicate or incorrect enrollments.

Examples:

- No duplicate course registration

- Accurate status updates after approvals
- Consistent report generation

NFR-4: Performance

The system shall provide fast response times for all operations to ensure smooth and uninterrupted user experience, especially during peak admission periods.

Examples:

- Quick dashboard refresh
- Fast search for student records
- No lag during bulk uploads

NFR-5: Scalability

The system shall support future growth in the number of students, courses, departments, and users without degradation in performance or functionality. It shall allow easy addition of new modules and features.

Examples:

- Add new departments without redesign
- Handle increased admissions annually
- Extend to mobile or portal integration

NFR-6: Availability

The system shall be continuously accessible to users during institutional working hours and must minimize downtime. The platform shall support high availability for academic operations.

Examples:

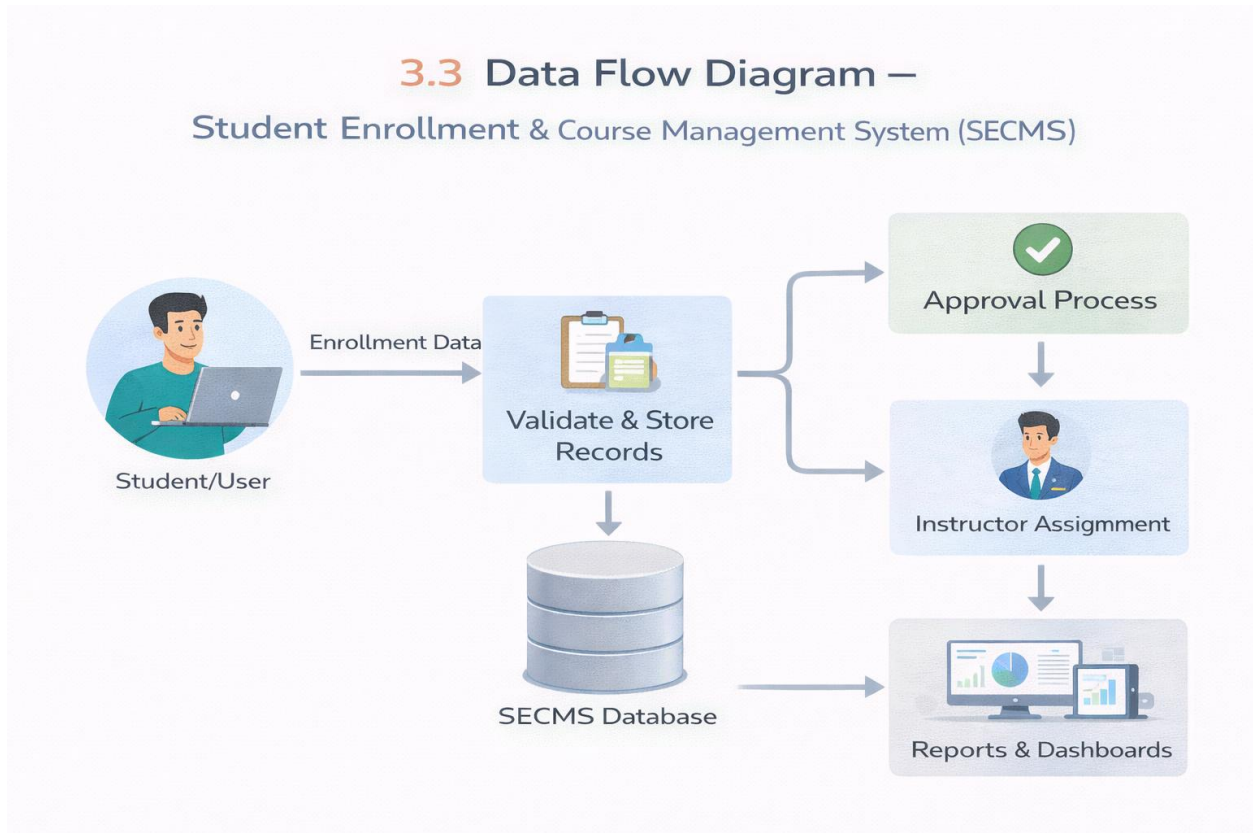
- 24/7 cloud access
- Reliable during peak admission season
- Automatic backup of student records

3.3 Data Flow Diagram

- User inputs student and enrollment data

- System validates and stores records
- Approval process routes data
- Reports and dashboards consume processed data.

The system processes enrollment requests, validates data, routes approvals, assigns instructors, and generates reports.



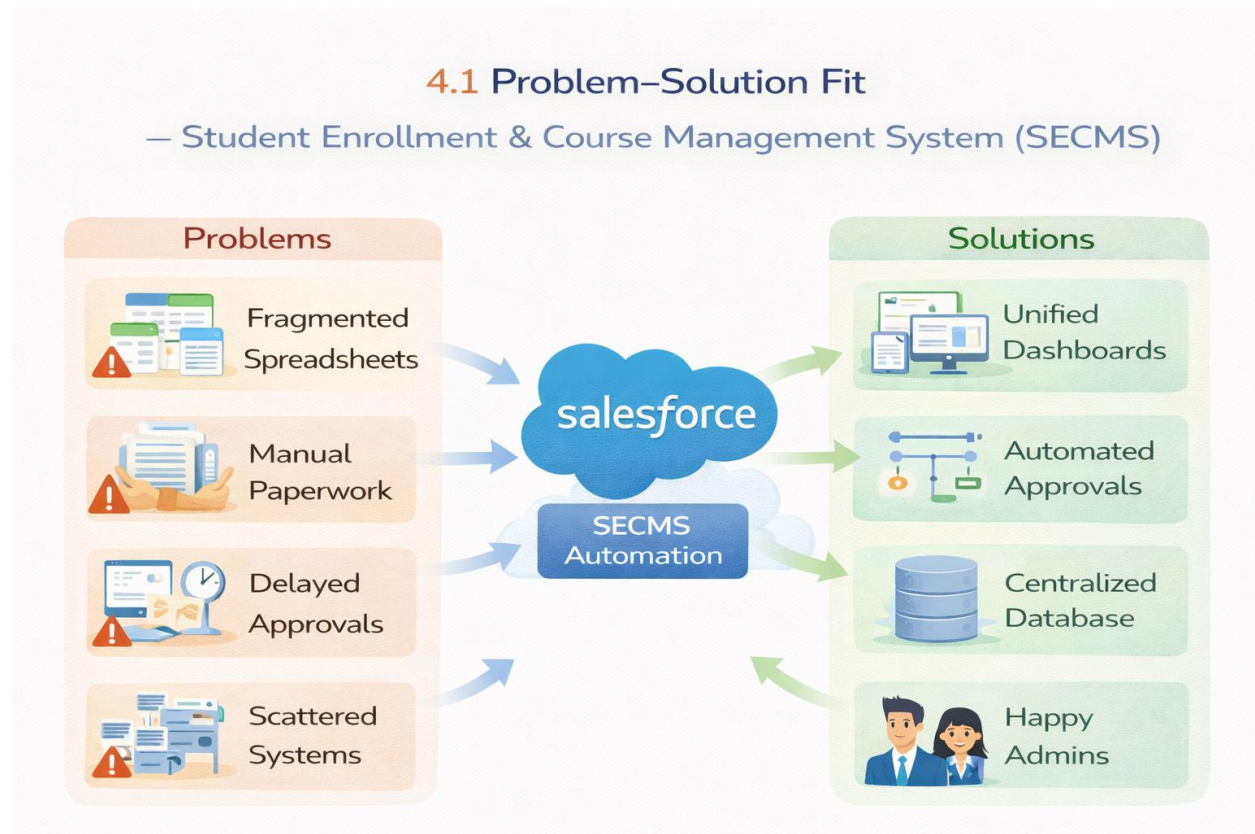
3.4 Technology Stack

Layer	Technology
UI	Salesforce Lightning
Logic	Salesforce Flows
Database	Salesforce Objects
Automation	Validation Rules, Approval Process
Reporting	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

SECMS directly addresses fragmented enrollment workflows by aligning user behavior, constraints, and automation using Salesforce-native capabilities.



4.2 Proposed Solution

Functional Requirements

Parameter	Description
Problem	Manual enrollment handling
Solution	Salesforce-based SECMS

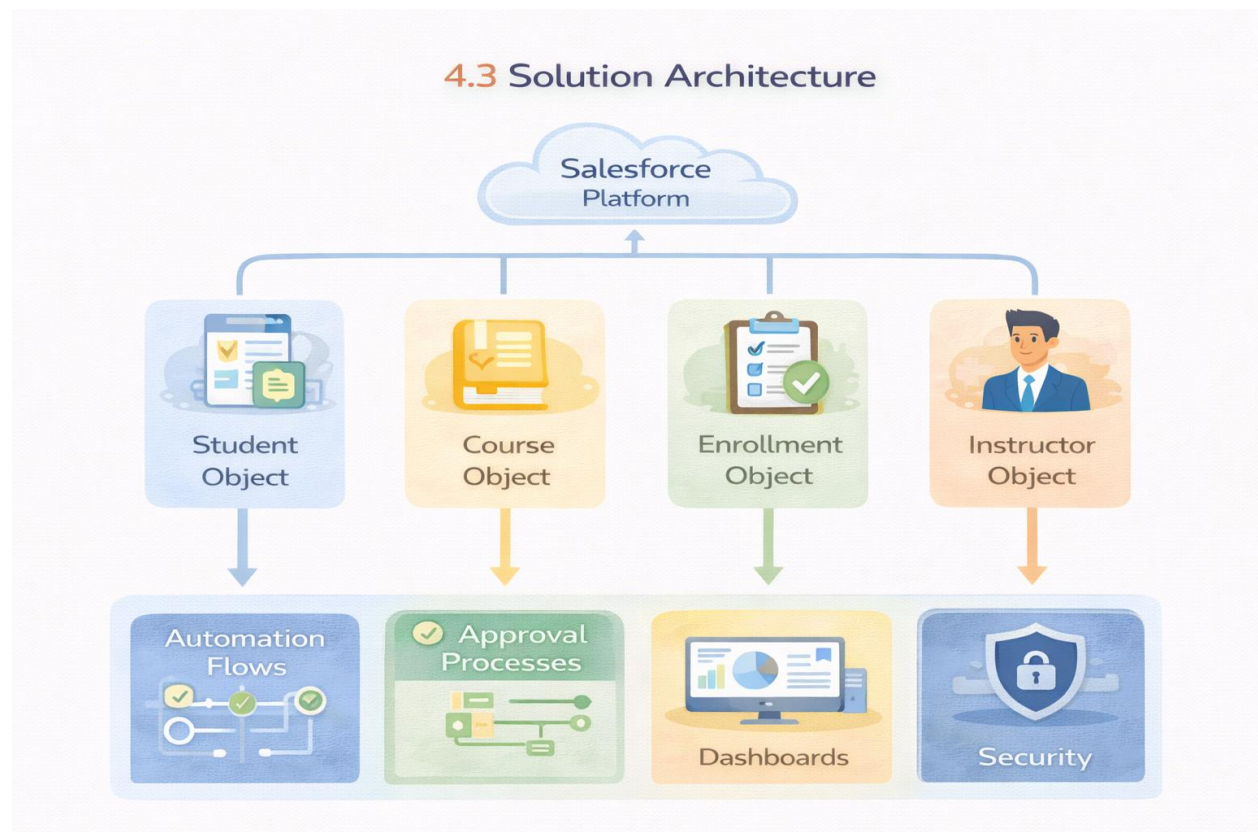
Novelty	End-to-end automation
Social Impact	Improved student experience
Revenue Model	Institutional deployment
Scalability	Multi-department support

4.3 Solution Architecture

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create a Salesforce developer account so that I can configure and deploy the Student Enrollment system..	1	High	Member1
Sprint-1	Data Modeling	USN-2	As an admin, I want to create custom objects for Student, Course, Enrollment, and Instructor so that academic data can be stored and managed efficiently.	3	High	Member2
Sprint-2	Data Modeling	USN-3	As a user, I want dedicated tabs for each object so that I can easily navigate between students, courses, enrollments, and instructors.	3	High	Member3
Sprint-2	Data Modeling	USN-4	As an admin, I want to create relevant fields such as student details, course information, and enrollment status so that complete records can be captured accurately.	5	High	Member4
Sprint-2	Data Modeling	USN-5	As an enrollment officer, I want a centralized Lightning App so that I can access all academic operations from one interface.	3	High	Member4
Sprint-2	Data Modeling	USN-6	As an admin, I want to define relationships between students, courses, and enrollments so that the system can logically link and track academic activities.	5	Medium	Member1
Sprint-2	Data Modeling	USN-7	As a user, I want customized page layouts so that only relevant information is displayed, improving clarity and ease of data entry.	5	High	Member2
Sprint-3	Automation	USN-8	As a system, I want to validate input data such as mandatory fields and enrollment limits so that incorrect or incomplete records are prevented.	3	High	Member4
Sprint-3	Automation	USN-9	As a system, I want automated flows to handle enrollment	3	High	Member4

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
			processing and notifications so that manual effort is reduced and processes become faster.			
Sprint-3	Automation	USN-10	As an administrator, I want an approval process for enrollment requests so that registrations are verified before confirmation.	5	High	Member1
Sprint-4	Security	USN-11	As an admin, I want to configure role-based security so that users can only access data relevant to their responsibilities.	5	High	Member2
Sprint-5	Reports	USN-12	As a manager, I want reports on student enrollments, course registrations, and instructor assignments so that I can monitor academic operations.	4	High	Member3
Sprint-6	Dashboards	USN-13	As a manager, I want visual dashboards showing enrollment statistics and trends so that I can quickly analyze system performance.	4	Medium	Member4

The architecture integrates Salesforce objects, automation flows, approval processes, dashboards, and security layers.



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		

6. PROJECT DEVELOPMENT PHASE – DETAILED IMPLEMENTATION

Milestone 1: Salesforce Developer Account Creation

The screenshot shows a web browser window with the URL `salesforce.com/products/free-trial/developer/`. The page has a dark blue header with the text "Build for the Agentic Enterprise. Sign up for your Developer Edition." Below this, a list of features is shown with checkmarks: "Build AI agents with Agentforce, grounded in your data with Data 360", "Connect Claude Code, Cursor, or Codex to your org with hosted MCP", "Build in the browser with the Agentforce Vibes IDE - no setup required", "Drag-and-drop when you want it, Apex when you need it", and "60+ MCP tools and 30+ coding skills, ready out of the box". On the right, a white sign-up form titled "Sign up for your Developer Edition" is displayed. It includes fields for First name (Dharshini), Last name (p), Job title (Student), Company (Dr. M. G. R. Government A...), Country/region (India), and Work email (pdharshu265@gmail.com). Each field has a green checkmark indicating it is valid. Below the form, there is a checkbox for agreeing to the Main Services Agreement, Developer Services, Salesforce Program Agreement, and API Terms of Service. A blue "Sign me up" button is at the bottom of the form. The browser's taskbar at the bottom shows the system clock as 07:03 PM on 25-09-2026, with a battery level of 54%.

Build for the Agentic Enterprise. Sign up for your Developer Edition.

The entire Salesforce platform - exposed as APIs, MCP tools, and CLI commands. Build apps, ship agents, and let coding agents do the heavy lifting with Salesforce Headless 360.

- ✓ Build AI agents with Agentforce, grounded in your data with Data 360
- ✓ Connect Claude Code, Cursor, or Codex to your org with hosted MCP
- ✓ Build in the browser with the Agentforce Vibes IDE - no setup required
- ✓ Drag-and-drop when you want it, Apex when you need it
- ✓ 60+ MCP tools and 30+ coding skills, ready out of the box

Sign up for your Developer Edition

A free Salesforce Platform environment with Agentforce and Data 360

First name: Dharshini ✓ Last name: p ✓

Job title: Student ✓ Company: Dr. M. G. R. Government A... ✓

Country/region: India ✓ Work email: pdharshu265@gmail.com ✓

☒ I agree to the [Main Services Agreement – Developer Services, Salesforce Program Agreement](#), and [API Terms of Service](#)

Your free trial may be provisioned on or migrated to Hyperforce, Salesforce's public cloud infrastructure.

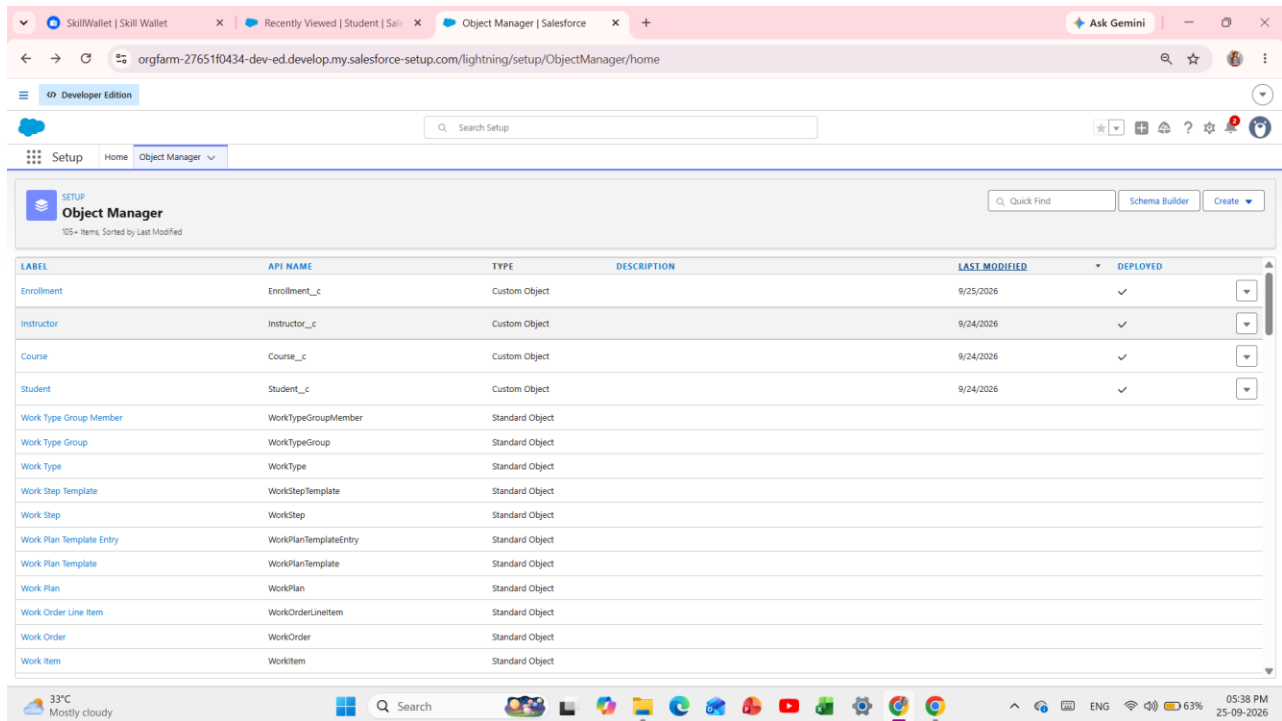
I acknowledge, as described in the Developer Documentation: (1) the Developer Edition includes autonomous and other generative AI features; and (2) Salesforce may limit use of those features and the org, and may terminate any org that has been inactive for 45 days.

We value your privacy. To learn more, visit our [Privacy Statement](#).

Sign me up

- Go to `https://developer.salesforce.com/signup`.
- Fill First Name, Last Name, Email, Role: Developer, Company: College Name, Country: India, and Username in the format `yourname@college.com`.

Milestone 2: Custom Object Creation



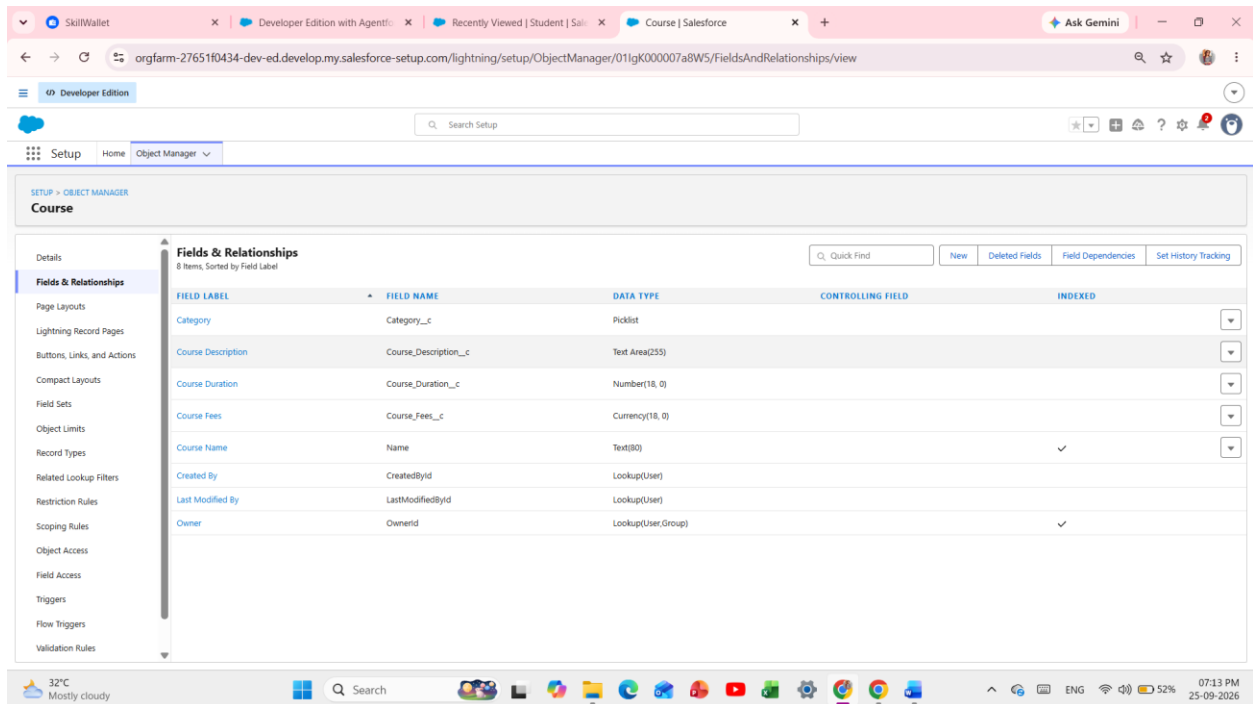
The screenshot shows the Salesforce Object Manager interface. The top navigation bar includes 'Setup', 'Home', and 'Object Manager'. The main content area displays a table of objects. The table has columns for LABEL, API NAME, TYPE, DESCRIPTION, LAST MODIFIED, and DEPLOYED. The first four rows show custom objects: Enrollment, Instructor, Course, and Student, all of type 'Custom Object' and created on 9/24/2026. The remaining rows show standard objects like Work Type Group Member, Work Type Group, Work Type, Work Step Template, Work Step, Work Plan Template Entry, Work Plan Template, Work Plan, Work Order Line Item, Work Order, and Work Item, all of type 'Standard Object'.

LABEL	API NAME	TYPE	DESCRIPTION	LAST MODIFIED	DEPLOYED
Enrollment	Enrollment__c	Custom Object		9/25/2026	✓
Instructor	Instructor__c	Custom Object		9/24/2026	✓
Course	Course__c	Custom Object		9/24/2026	✓
Student	Student__c	Custom Object		9/24/2026	✓
Work Type Group Member	WorkTypeGroupMember	Standard Object			
Work Type Group	WorkTypeGroup	Standard Object			
Work Type	WorkType	Standard Object			
Work Step Template	WorkStepTemplate	Standard Object			
Work Step	WorkStep	Standard Object			
Work Plan Template Entry	WorkPlanTemplateEntry	Standard Object			
Work Plan Template	WorkPlanTemplate	Standard Object			
Work Plan	WorkPlan	Standard Object			
Work Order Line Item	WorkOrderLineItem	Standard Object			
Work Order	WorkOrder	Standard Object			
Work Item	WorkItem	Standard Object			

Fig 1: Custom Objects Created

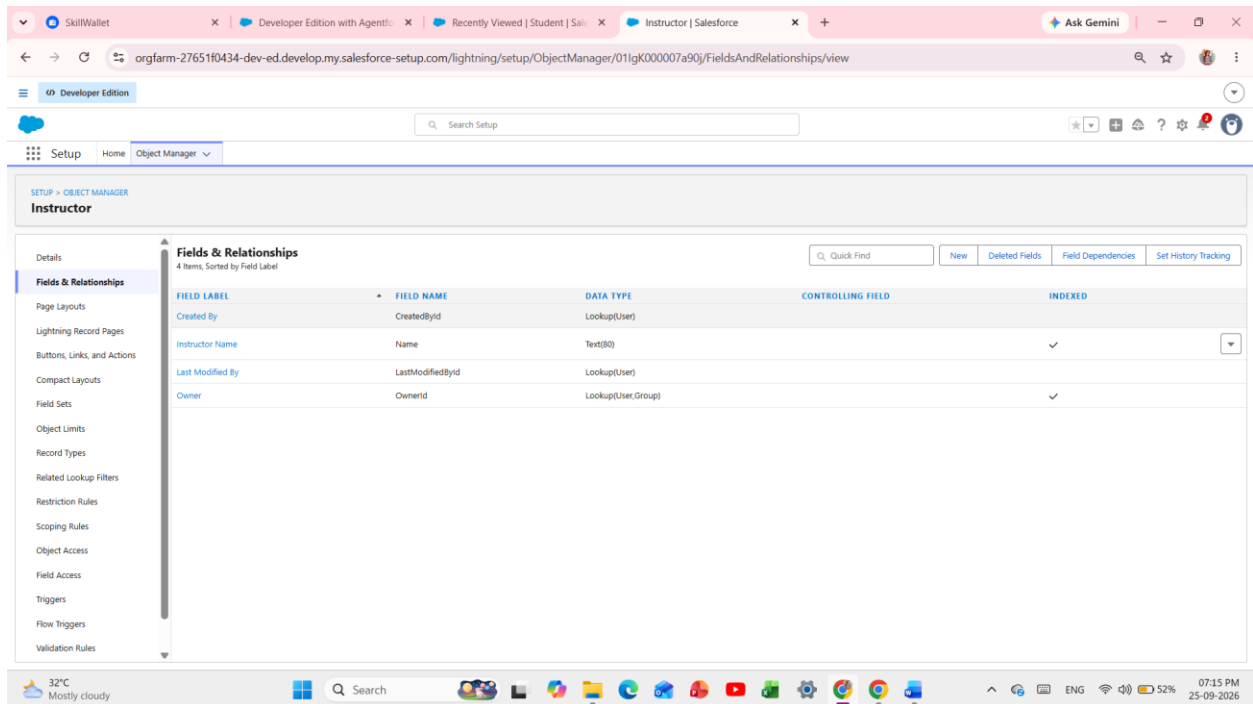
- Setup → Object Manager → Create → Custom Object.
- Label: Student; Plural: Students; API: Student__c; Record Name: Student Name (Text); enable Allow Reports and Allow Search.
- Similarly create Course__c, Instructor__c, and Enrollment__c.

Milestone 3: Fields Creation



This screenshot shows the Salesforce Setup interface for the 'Course' object. The 'Fields & Relationships' section is active, displaying a table of 8 fields. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, and others. The top of the page shows the browser tabs and the Salesforce URL.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Category	Category__c	Picklist		
Course Description	Course_Description__c	Text Area(255)		
Course Duration	Course_Duration__c	Number(18, 0)		
Course Fees	Course_Fees__c	Currency(18, 0)		
Course Name	Name	Text(80)		✓
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓



This screenshot shows the Salesforce Setup interface for the 'Instructor' object. The 'Fields & Relationships' section is active, displaying a table of 4 fields. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, and others. The top of the page shows the browser tabs and the Salesforce URL.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Instructor Name	Name	Text(80)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓

SkillWallet

Developer Edition with Agenti...

Recently Viewed | Student | Sal...

Student | Salesforce

Ask Gemini

orgfarm-27651f0434-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/01lgK000007a7wb/FieldsAndRelationships/view

Developer Edition

Setup Home Object Manager

SETUP > OBJECT MANAGER

Student

Details

Fields & Relationships

9 Items, Sorted by Field Label

Quick Find

New

Deleted Fields

Field Dependencies

Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Date of birth	Date_of_birth__c	Date		
Email	Email__c	Email		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Phone	Phone__c	Phone		
Student Address	Student_Address__c	Text Area(255)		
Student Name	Name	Text(80)		✓
Student Status	Student_status__c	Picklist		

32°C Mostly cloudy

Search

ENG 53% 07:11 PM 25-09-2026

SkillWallet

Developer Edition with Agenti...

Recently Viewed | Student | Sal...

Enrollment | Salesforce

Ask Gemini

orgfarm-27651f0434-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/01lgK000007a8sf/FieldsAndRelationships/view

Developer Edition

Setup Home Object Manager

SETUP > OBJECT MANAGER

Enrollment

Details

Fields & Relationships

12 Items, Sorted by Field Label

Quick Find

New

Deleted Fields

Field Dependencies

Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Course	Course__c	Lookup(Course)		✓
Created By	CreatedById	Lookup(User)		
Enrollment Date	Enrollment_Date__c	Date		
Enrollment Name	Name	Text(80)		✓
Enrollment Status	Enrollment_Status__c	Picklist		
Fees Paid	Fees_Paid__c	Checkbox		
Instructor	Instructor__c	Lookup(Instructor)		✓
Instructor User	Instructor_User__c	Lookup(User)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Record Type	RecordTypeId	Record Type		✓
Student	Student__c	Lookup(Student)		✓

32°C Mostly cloudy

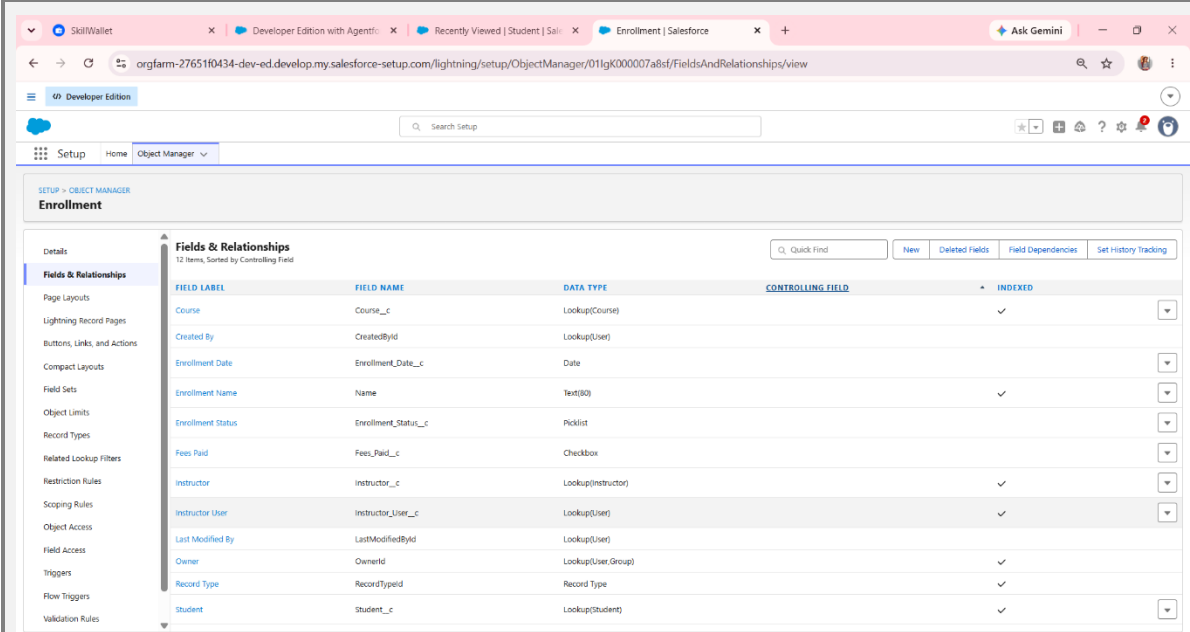
Search

ENG 52% 07:12 PM 25-09-2026

Object fields / Fields & Relationships showing the configured Student, Course, Instructor and Enrollment fields

- Student__c: Email (Email), Phone (Phone), DOB (Date), Status (Picklist: Active, Inactive, Graduated), Address (Long Text Area).
- Course__c: Category (Picklist: Technical, Language, Non-Technical), Duration (Number), Course Fee (Currency), Description (Long Text).
- Instructor__c: Instructor Code (Text: INSTR_A, INSTR_B, INSTR_C), Expertise (Picklist), Phone, Email.
- Enrollment__c: Student (Lookup Student__c), Course (Lookup Course__c), Instructor (Lookup Instructor__c), Instructor User (Lookup User – Critical), Enrollment Date (Date), Enrollment Status (Picklist: Requested, Approved, Rejected), Fees Paid (Checkbox), Total Amount (Currency).

Milestone 4: Relationships



The screenshot shows the Salesforce Setup interface for the 'Enrollment' object. The 'Fields & Relationships' section is active, displaying a table of 12 fields. The table columns are FIELD LABEL, FIELD NAME, DATA TYPE, CONTROLLING FIELD, and INDEXED. The fields are sorted by Controlling Field.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Course	Course__c	Lookup(Course)		✓
Created By	CreatedById	Lookup(User)		
Enrollment Date	Enrollment_Date__c	Date		
Enrollment Name	Name	Text(80)		✓
Enrollment Status	Enrollment_Status__c	Picklist		
Fees Paid	Fees_Paid__c	Checkbox		
Instructor	Instructor__c	Lookup(Instructor)		✓
Instructor User	Instructor_User__c	Lookup(User)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User Group)		✓
Record Type	RecordTypeId	Record Type		✓
Student	Student__c	Lookup(Student)		✓

Enrollment lookup relationships to Student, Course and Instructor
(Paste the Salesforce screenshot here)

- Relationships are created through the Lookup fields defined in Milestone 3.

Milestone 5: Validation Rules

The first screenshot shows the Salesforce Object Manager interface for an 'Enrollment Validation Rule'. The rule is named 'Fees_Must_Be_Paid_Before_Approval' and is active. The error condition formula is: `AND( ISPICKVAL( Enrollment_Status__c, 'Approved'), NOT( Fees_Paid__c = TRUE ) )`. The error message is 'Fees must be paid before approval'. The error location is 'Top of Page'. The rule was created by 'Chandana P' on 9/25/2026 at 6:59 AM.

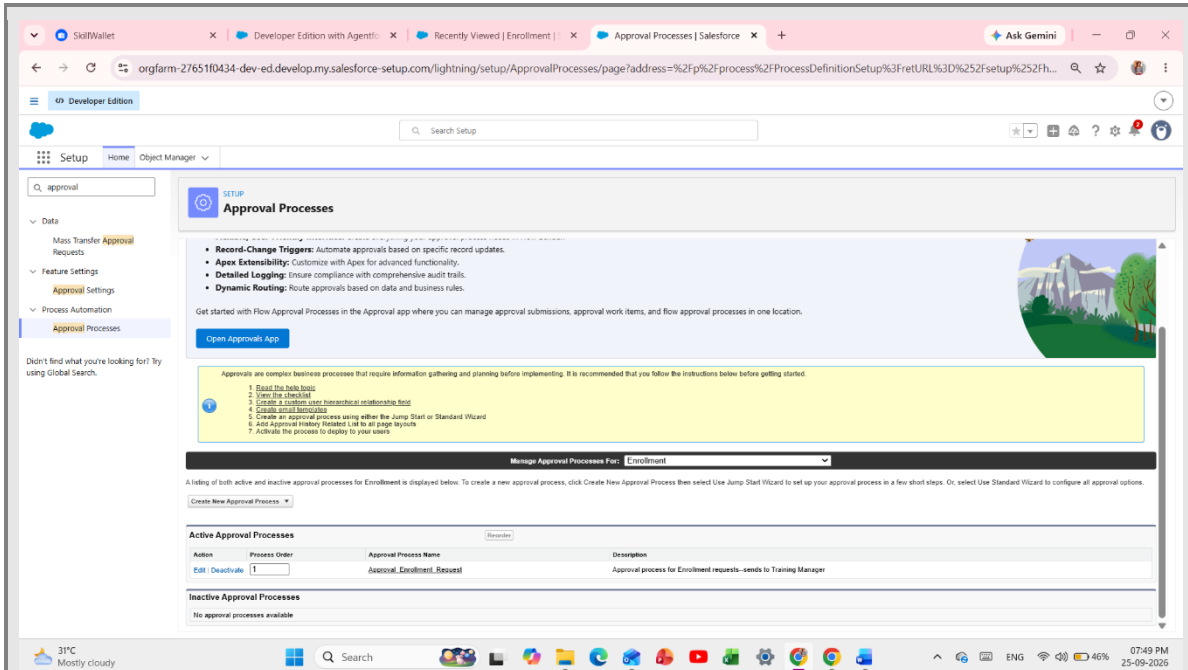
The second screenshot shows a 'New Enrollment: New Enrollment' form in Salesforce. The form contains the following fields: Enrollment Name (Test Enroll 02), Student (Kannan), Course (Salesforce Admin), Instructor (INSTR_B), Enrollment Date (9/25/2026), Enrollment Status (Approved), Fees Paid (checkbox), and Instructor User (Search People...). A red error message is displayed: 'We hit a snag. Review the errors on this page. Fees must be paid before approval.' The error message is in red text and has a red border.

Validation Rule error showing “Fees must be paid before approval” in red

Fig 2: Validation Rule Testing

- Rule 1 – Student_Email_Domain_Validation: Object Student__c → Validation Rules → New. Formula: NOT(CONTAINS(Email__c, "@student.edu")). Error: Email must be valid student institutional email. Location: Email field.
- Rule 2 – Fees_Must_Be_Paid_Before_Approval: Object Enrollment__c. Formula: AND(ISPICKVAL(Enrollment_Status__c, "Approved"), Fees_Paid__c = FALSE). Error: Fees must be paid before approving enrollment.

Milestone 6: Approval Process



The screenshot shows the Salesforce Setup interface for Approval Processes. The left sidebar contains navigation links for Data, Feature Settings, and Process Automation. The main content area displays the 'Approval Processes' setup page for the 'Enrollment' object. It includes a 'Manage Approved Processes For: Enrollment' dropdown, a 'Create New Approval Process' button, and a table of 'Active Approval Processes'. The table shows one active process: 'Approval_Enrollment_Request' with a process order of 1. Below the table, it states 'Inactive Approval Processes: No approval processes available'.

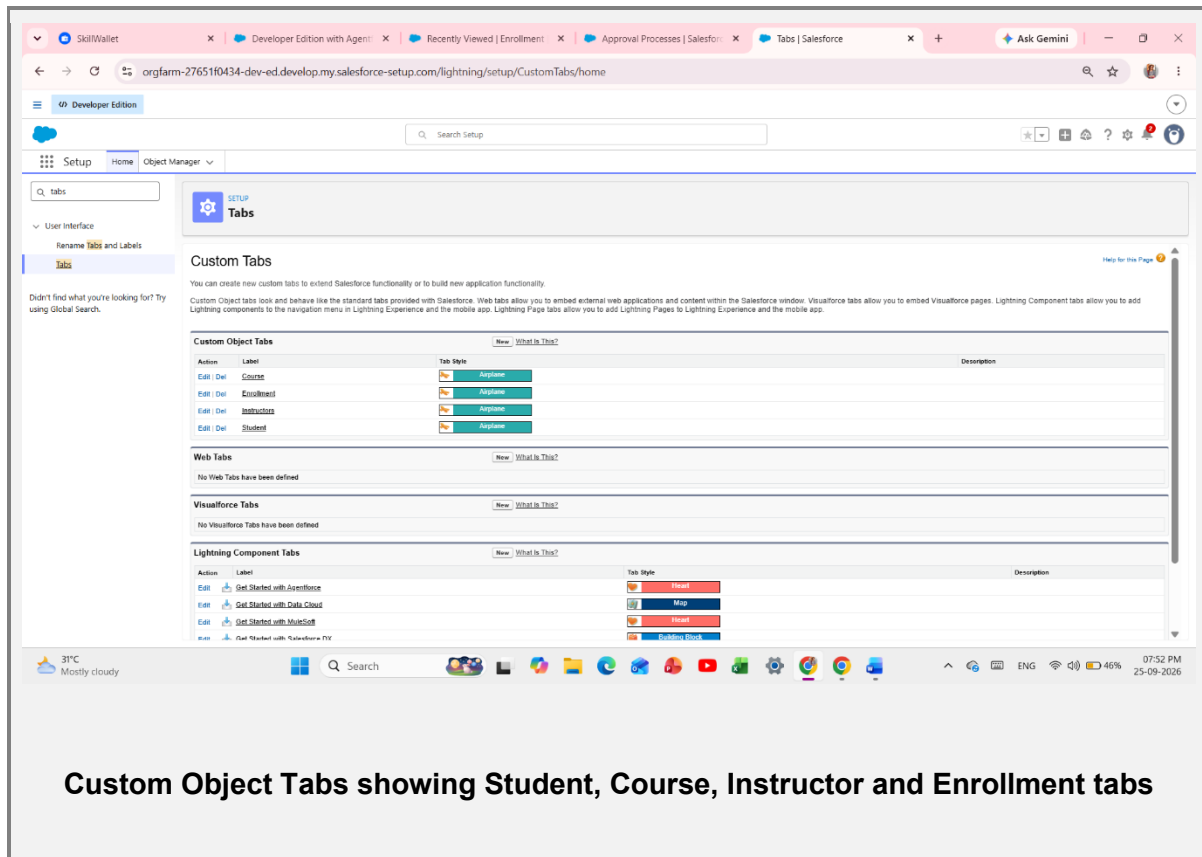
Approval Process page showing the enrollment approval process is Active

[Insert screenshot here]

Fig 3: Approval Process Active

- Setup → Approval Processes → Select Enrollment__c → Create New Approval Process → Use Standard Setup.
- Name: Approval_Enrollment_Request. Entry Criteria: Enrollment_Status__c = Requested. Approver: Training Manager User.
- Final Approval Action: Field Update → Enrollment_Status__c = Approved.
- Final Rejection Actions: Field Update → Enrollment_Status__c = Rejected + Email Alert to Student.
- Activate the approval process.

Milestone 7: Tabs



The screenshot shows the Salesforce Setup interface for Custom Tabs. The left sidebar has a search bar with 'tabs' entered and a list of tabs including 'User Interface' and 'Tabs'. The main content area is titled 'Custom Tabs' and includes a 'New' button. Below this, there are four sections: 'Custom Object Tabs', 'Web Tabs', 'Visualforce Tabs', and 'Lightning Component Tabs'. The 'Custom Object Tabs' section contains a table with four rows: 'Course', 'Enrollment', 'Instructor', and 'Student'. Each row has columns for 'Action', 'Label', 'Tab Style', and 'Description'. The 'Tab Style' column shows 'Appspace' for all four tabs. The 'Web Tabs', 'Visualforce Tabs', and 'Lightning Component Tabs' sections each have a 'New' button and a message stating 'No [type] Tabs have been defined'.

Custom Object Tabs

Action	Label	Tab Style	Description
Edit Del	Course	Appspace	
Edit Del	Enrollment	Appspace	
Edit Del	Instructor	Appspace	
Edit Del	Student	Appspace	

Web Tabs

No Web Tabs have been defined

Visualforce Tabs

No Visualforce Tabs have been defined

Lightning Component Tabs

Action	Label	Tab Style	Description
Edit	Get Started with ApexFlow	Mobile	
Edit	Get Started with Data Cloud	Mobile	
Edit	Get Started with MuleSoft	Mobile	
Edit	Get Started with Salesforce PIV	Mobile	

Custom Object Tabs showing Student, Course, Instructor and Enrollment tabs

- Setup → Tabs → Custom Object Tabs → New → Select Student → Tab Style → Next → Add to all Profiles → Save.
- Repeat for Course, Instructor, and Enrollment.

Milestone 8: Lightning App

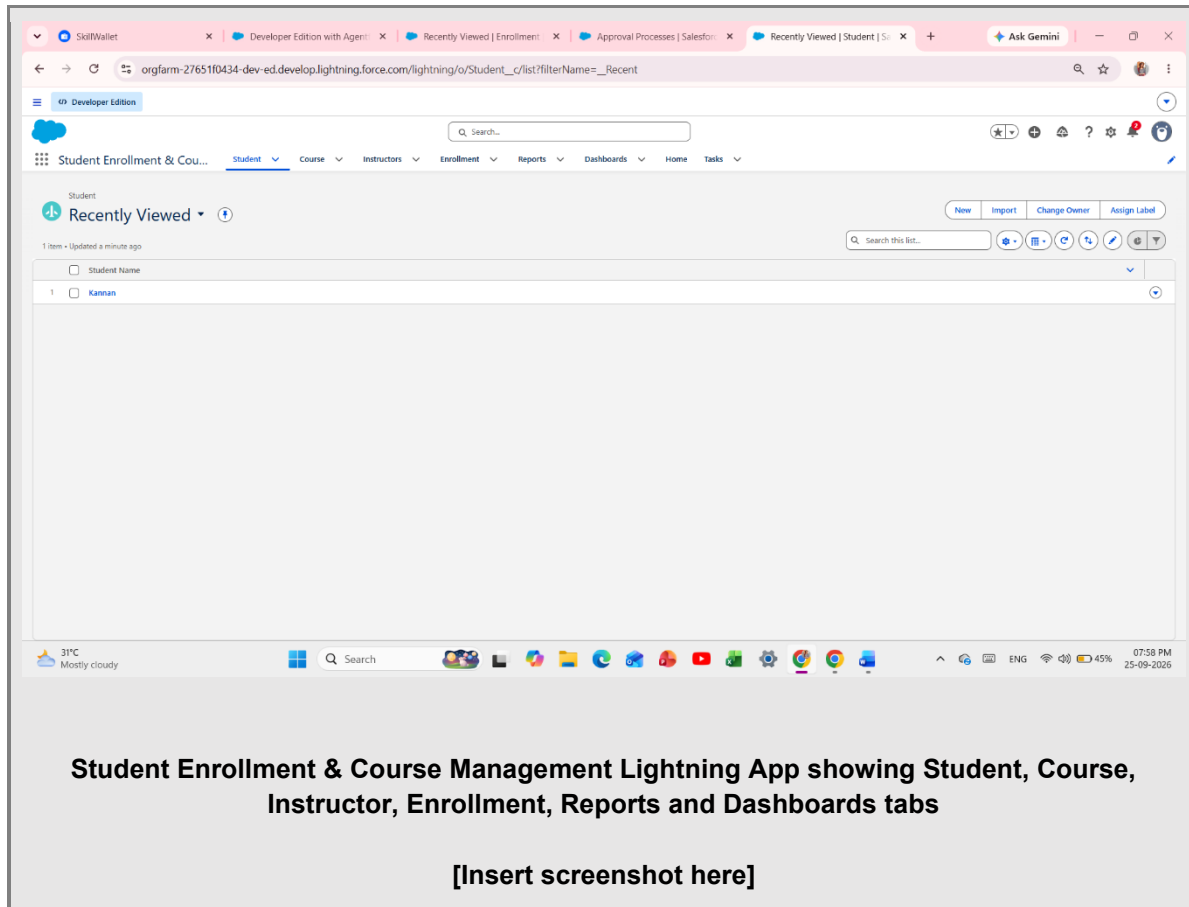


Fig 4: Lightning App with Tabs

- Setup → App Manager → New Lightning App.
- App Name: Student Enrollment & Course Management. Navigation: Standard. Profiles: System Admin.
- Add Tabs: Student, Course, Instructor, Enrollment, Reports, Dashboards, Home → Save & Finish.

Milestone 9: Page Layouts

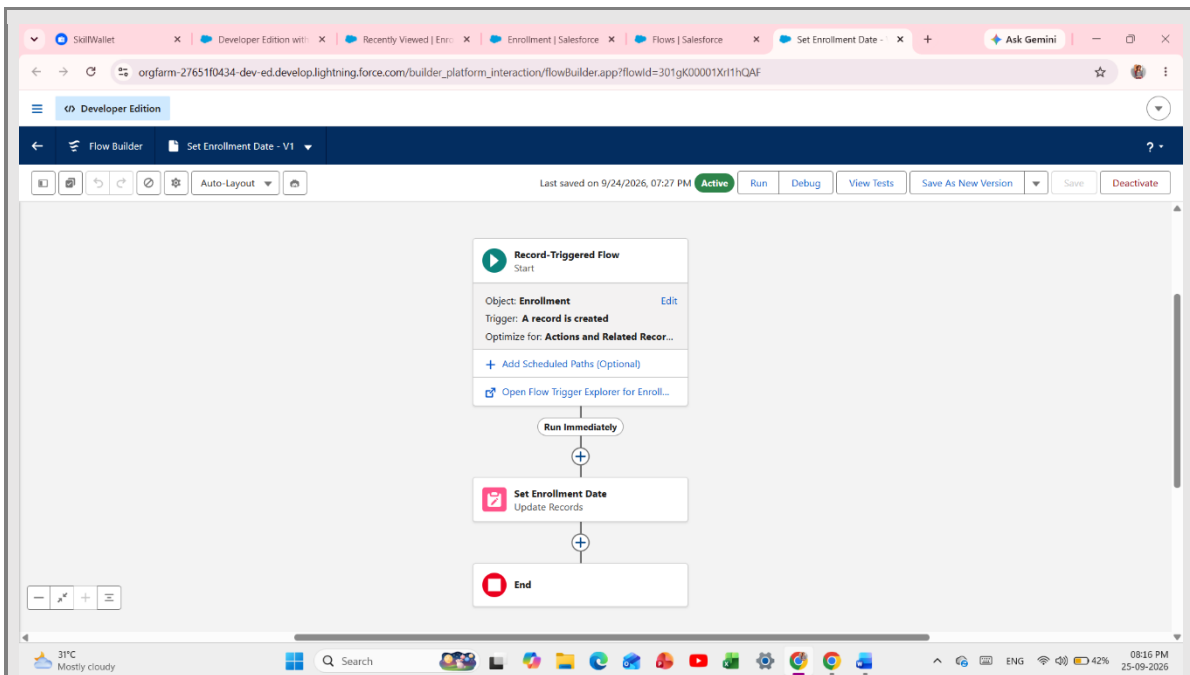
The first screenshot shows the 'Student' object page layout configuration in Salesforce Developer Edition. The left sidebar lists various configuration options, with 'Page Layouts' selected. The main area displays the 'Layout Properties' dialog, showing fields like 'Student Address', 'Student Name', 'Student Status', and 'Phone'. Below this, there are sections for 'Mobile Cards (Salesforce mobile only)', 'Related Lists', 'Open Activities', 'Activity History', and 'Enrollment'. The 'Related Lists' section shows a table with columns: Subject, Name, Task, Due Date, Status, Priority, and Assigned To. The 'Open Activities' section shows a table with columns: Subject, Name, Task, Due Date, Assigned To, and Last Modified Date/Time. The 'Enrollment' section shows a table with columns: Enrollment Name and Sample Text.

The second screenshot shows the 'Enrollment' object page layout configuration. The left sidebar lists various configuration options, with 'Record Types' selected. The main area displays the 'Record Types' configuration page, showing a table with columns: RECORD TYPE LABEL, DESCRIPTION, ACTIVE, and MODIFIED BY. The table lists two record types: 'New Enrollment' and 'Re-Enrollment', both of which are active and were modified by 'Dhanshi P.' on 9/24/2026 at 7:07 AM.

Enrollment Record Types and Page Layout / Related Lists configuration
(Paste the Salesforce screenshot here)

- Setup → Object Manager → Student__c → Page Layouts → Edit Layout → Related Lists → Drag Enrollments Related List → Save.
- Repeat for Course__c and Instructor__c.
- Create Record Types for Enrollment: New Enrollment and Re-Enrollment, assigning different page layouts.

Milestone 10: Flows – Most Important



The screenshot displays the Salesforce Flow Builder interface. The top navigation bar shows the 'Flow Builder' tab and the 'Set Enrollment Date - V1' flow. The flow canvas shows a 'Record-Triggered Flow' starting with the trigger 'Object: Enrollment' and 'Trigger: A record is created'. The flow steps include 'Run Immediately', 'Set Enrollment Date' (Update Records), and 'End'. The flow is marked as 'Active'.

Flows list showing the active automation flows, including SetEnrollmentDate, AutoAssign_Instructor, and Change_Owner_To_Instructor

[Insert screenshot here]

Fig 5: Flows – Auto Assign (Active)

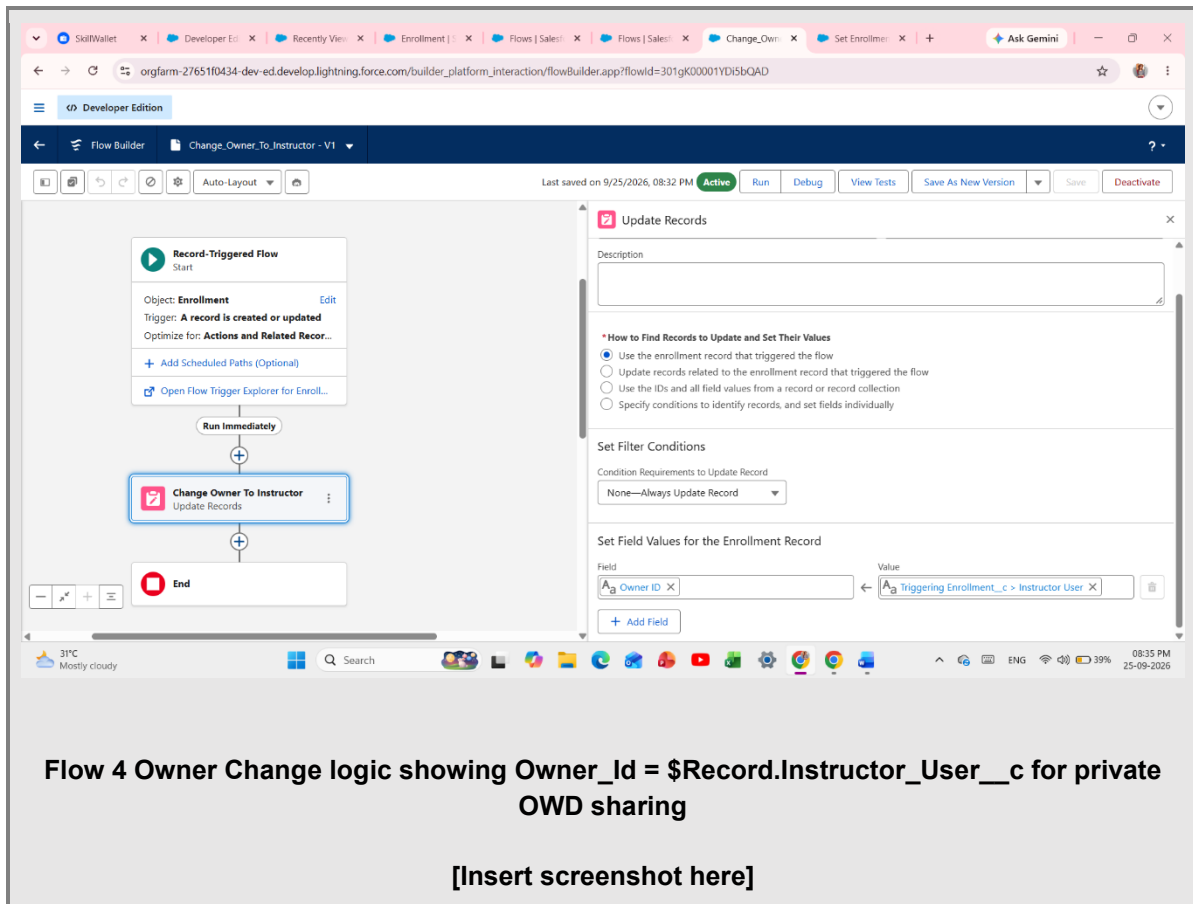
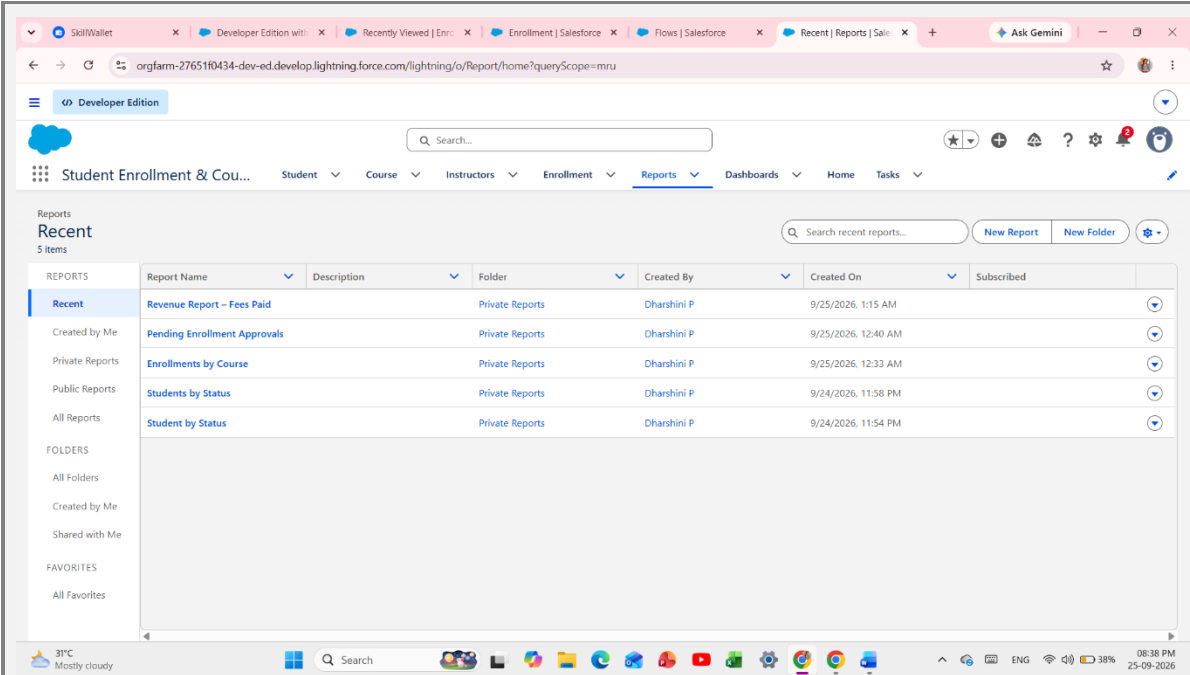


Fig 6: Owner Change Flow – Critical for Private OWD

- Flow 1 – Auto-Populate Enrollment Date: Record-Triggered Flow on Enrollment__c when a record is created → Update Records → Enrollment_Date__c = \$Flow.CurrentDate → Save as SetEnrollmentDate → Activate.
- Flow 2 – Send Email When Approved: Record-Triggered Flow on Enrollment__c update → Condition Status = Approved → Get Student__c → Email Alert → Update Student Status = Active → Activate.
- Flow 3 – Auto-Assign Instructor Based on Course Category: Trigger on Enrollment__c Created/Updated → Get Course__c → Decision by Course Category → Technical = INSTR_A; Language = INSTR_B; Non-Technical = INSTR_C → Update Enrollment Instructor__c with found Instructor Id → Activate.
- Flow 4 – Owner Change for Private Sharing: Trigger on Enrollment__c Created/Updated → Condition Instructor_User__c != null → Update OwnerId = \$Record.Instructor_User__c → Save as Change_Owner_To_Instructor → Activate. This makes Enrollment Private OWD work by allowing the assigned instructor to own and access the record.

Milestone 11: Reports



The screenshot shows the Salesforce Reports interface. The left sidebar contains navigation options: Reports, Recent (5 items), All Reports, FOLDERS (All Folders, Created by Me, Shared with Me), and FAVORITES (All Favorites). The main area displays a table of recent reports:

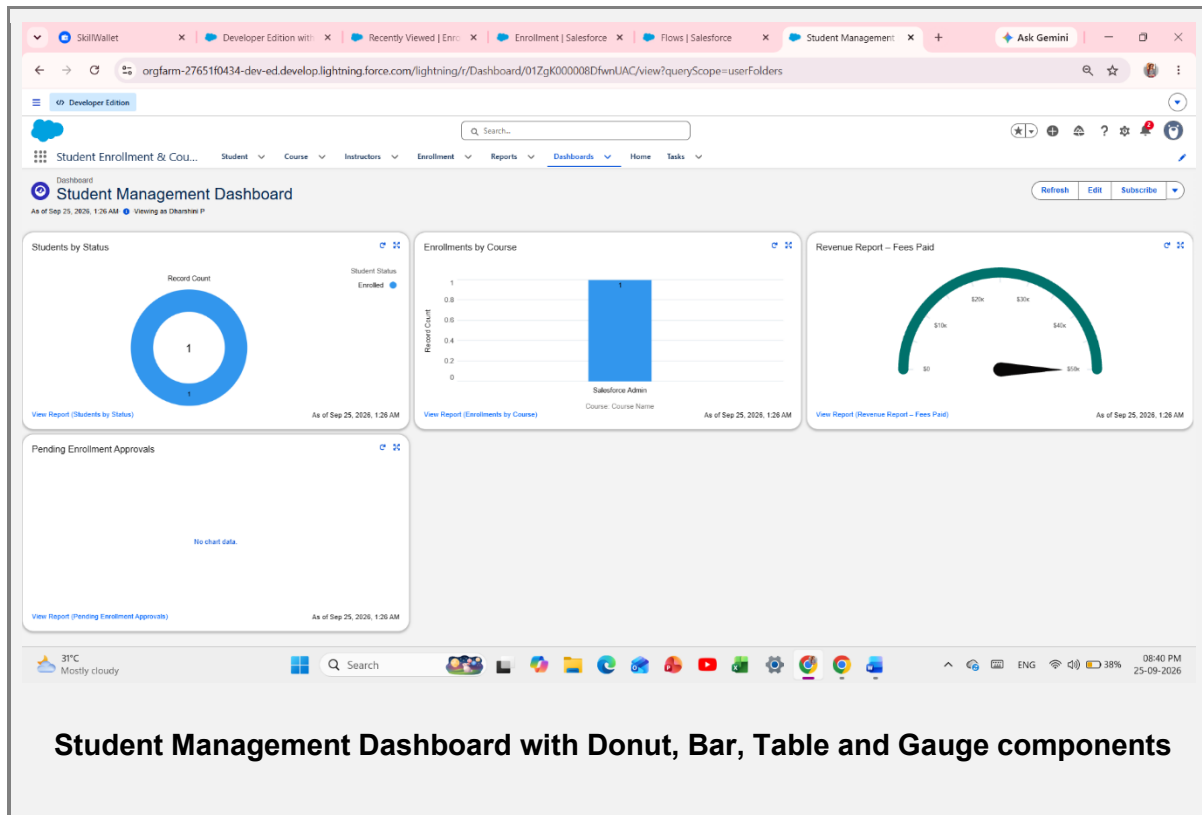
REPORTS	Report Name	Description	Folder	Created By	Created On	Subscribed
Recent	Revenue Report - Fees Paid		Private Reports	Dharshini P	9/25/2026, 1:15 AM	
Created by Me	Pending Enrollment Approvals		Private Reports	Dharshini P	9/25/2026, 12:40 AM	
Private Reports	Enrollments by Course		Private Reports	Dharshini P	9/25/2026, 12:33 AM	
Public Reports	Students by Status		Private Reports	Dharshini P	9/24/2026, 11:58 PM	
All Reports	Student by Status		Private Reports	Dharshini P	9/24/2026, 11:54 PM	

Below the table, there are sections for FOLDERS and FAVORITES. The bottom of the screenshot shows a Windows taskbar with the date 25-09-2026 and time 08:38 PM.

**Reports list showing Students by Status, Enrollments by Course, Pending Approvals and Revenue Report
(Paste the Salesforce screenshot here)**

- Students by Status: Reports → New Report → Students → Group by Status.
- Enrollments by Course: Reports → New Report → Enrollments → Group by Course Name.
- Pending Enrollment Approvals: Filter Status = Requested.
- Revenue Report: Filter Fees Paid = TRUE → Group by Course → Summarize Total Amount.

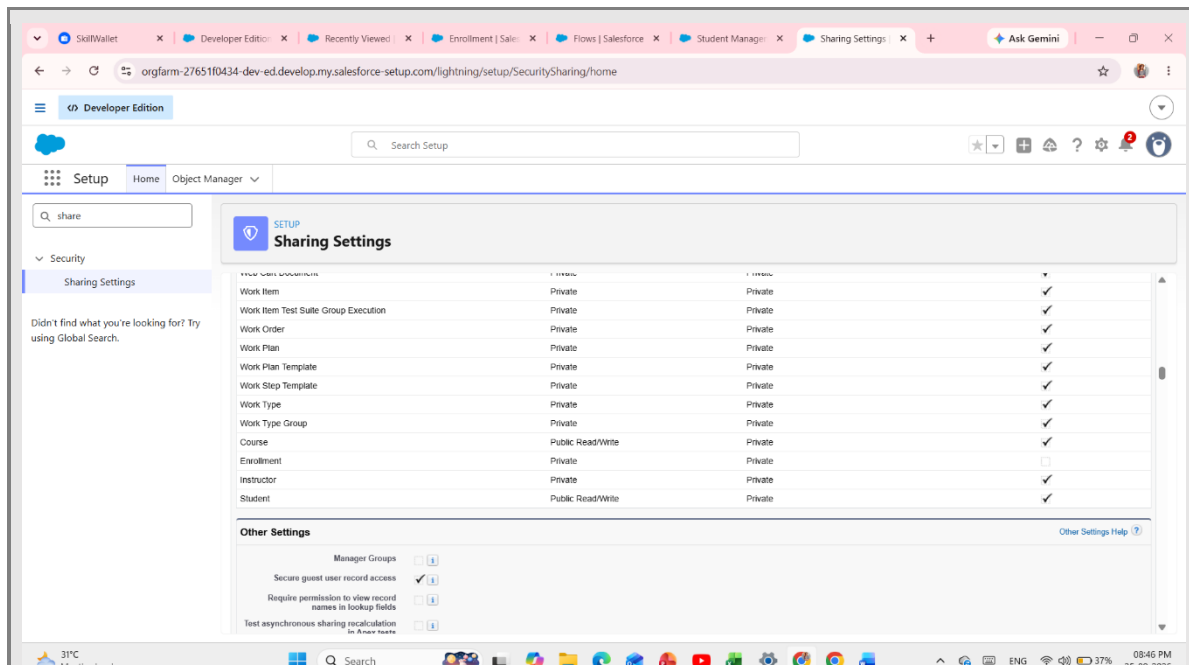
Milestone 12: Dashboards



Student Management Dashboard with Donut, Bar, Table and Gauge components

- Dashboards → New Dashboard → Name: Student Management Dashboard.
- Add components: Donut Chart from Students by Status, Bar Chart from Enrollments by Course, Table from Pending Approvals, and Gauge from Revenue Report → Save.

Milestone 13: Security Setup



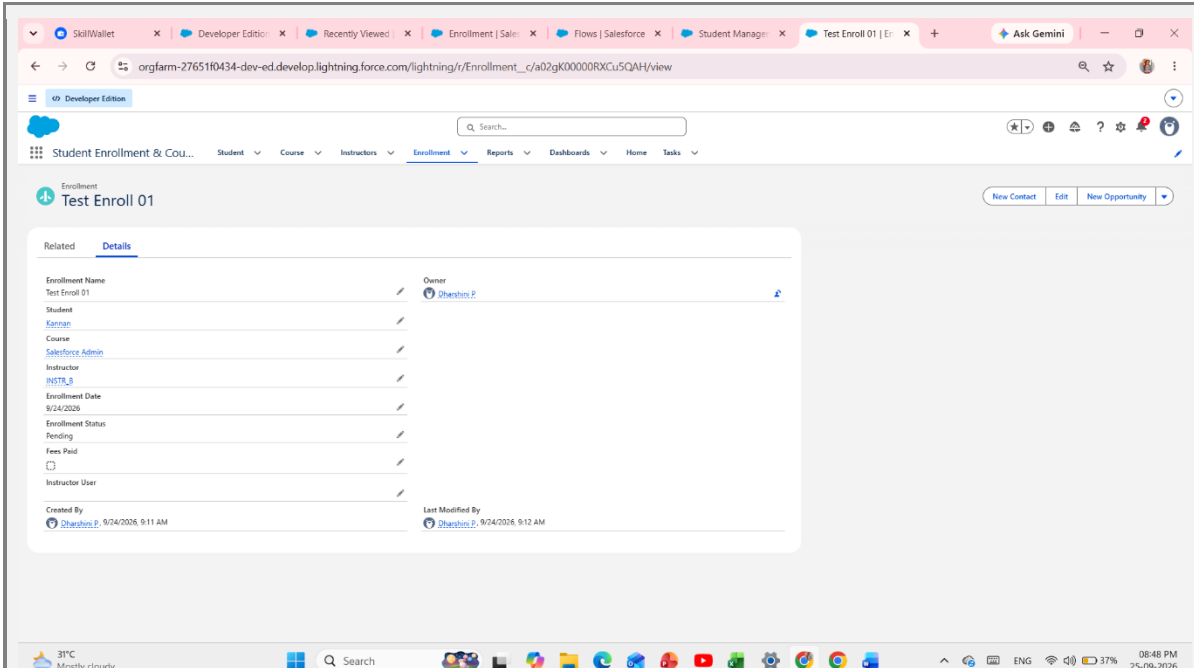
Setup → Sharing Settings showing Student__c = Public Read/Write, Course__c = Public Read/Write, Enrollment__c = Private

[Insert screenshot here]

Fig 7: Organization-Wide Defaults – Sharing Settings

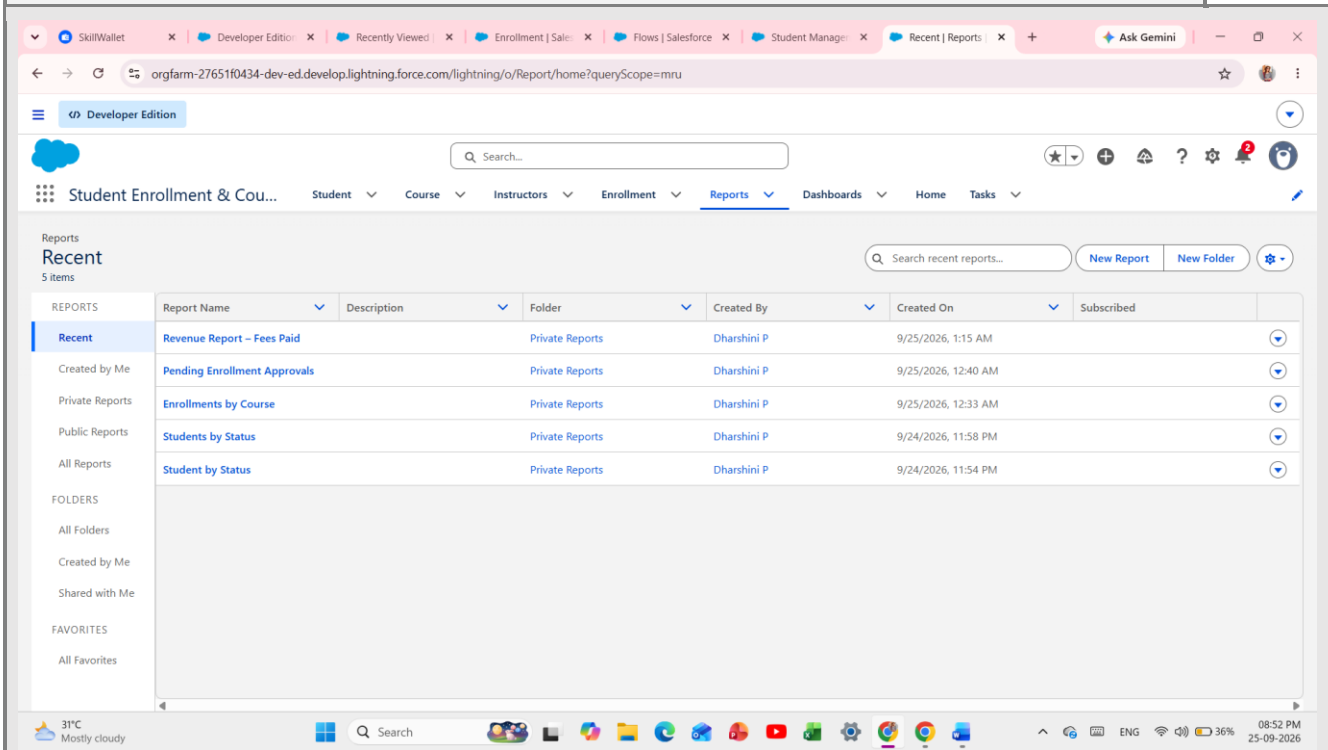
- Setup → Sharing Settings → Organization-Wide Defaults.
- Student__c: Public Read/Write – all users need to see students.
- Course__c: Public Read/Write – course catalog is public.
- Enrollment__c: Private – only the assigned instructor should see the enrollment, enforced through the Owner Change Flow.
- Profiles: Clone System Admin to create Enrollment Officer Profile with Enrollment Read/Create/Edit, Student Read, and Course Read. Instructor Profile: Enrollment Read-only.
- Permission Sets: Dashboard Editor and Custom Object Access – assign to appropriate users.

7. FUNCTIONAL & PERFORMANCE TESTING



The screenshot displays the Salesforce Developer Edition interface. The top navigation bar includes tabs for SkillWallet, Developer Edition, Recently Viewed, Enrollment | Sales, Flows | Salesforce, Student Manager, and Test Enroll 01 | Edit. The main content area shows the 'Test Enroll 01' record details. The 'Related' tab is active, showing a list of related records with fields like Enrollment Name, Student, Course, Instructor, Enrollment Date, Enrollment Status, Pending, Fees Paid, and Instructor User. The 'Details' tab is also visible, showing fields like Enrollment Name, Student, Course, Instructor, Enrollment Date, Enrollment Status, Pending, Fees Paid, and Instructor User. The record is created by Dharshini P. on 9/24/2026 at 9:11 AM and last modified by Dharshini P. on 9/24/2026 at 9:12 AM.

Overall Salesforce testing evidence: validation error, approval, flow, reports/dashboard or enrollment record proof



The screenshot displays the Salesforce Developer Edition interface, specifically the 'Recent Reports' page. The top navigation bar includes tabs for SkillWallet, Developer Edition, Recently Viewed, Enrollment | Sales, Flows | Salesforce, Student Manager, and Recent | Reports. The main content area shows a list of recent reports with columns for Report Name, Description, Folder, Created By, Created On, and Subscribed. The 'Recent' report is highlighted, showing details for 'Revenue Report - Fees Paid', 'Pending Enrollment Approvals', 'Enrollments by Course', 'Students by Status', and 'Student by Status'. The 'Enrollments by Course' and 'Students by Status' reports are the focus of the caption.

Reports page showing “Students by Status” and “Enrollments by Course” reports

Fig 8: Reports – Students by Status & Enrollments by Course

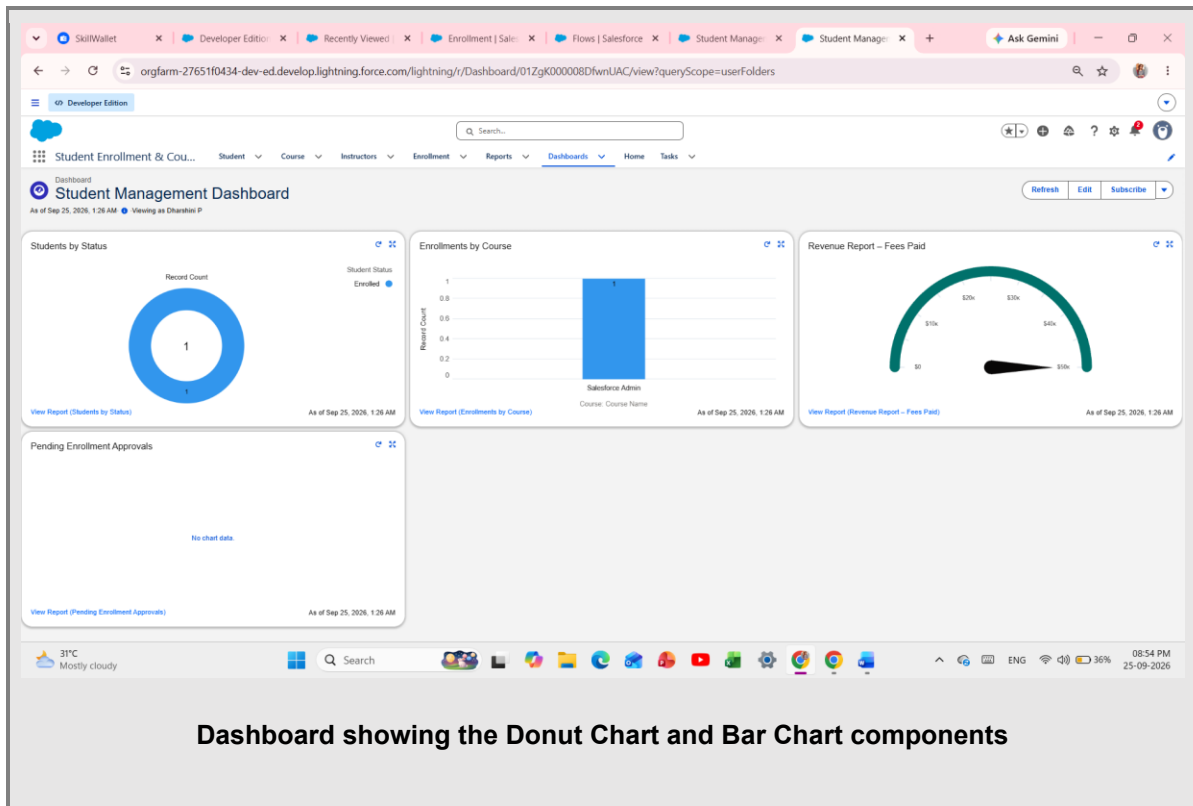


Fig 9: Student Management Dashboard

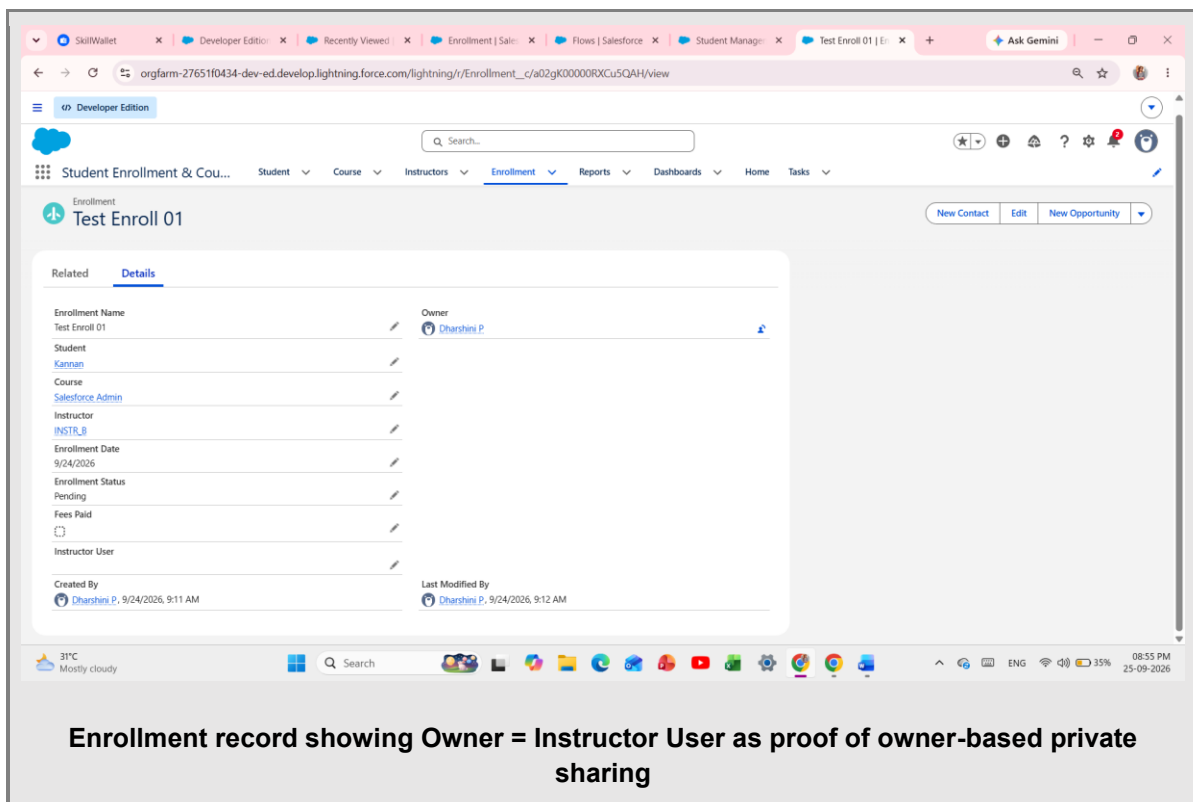


Fig 10: Enrollment Owner Proof – Instructor User

Test Case	Test	Expected Result	Result
TC-01	Create Student with invalid email	Validation error	PASS
TC-02	Create Enrollment without Fees Paid and Approve	Validation error	PASS
TC-03	Create Enrollment with Status Requested	Approval process triggered	PASS
TC-04	Create Enrollment	Enrollment Date auto-filled today	PASS
TC-05	Create Enrollment with Technical Course	Instructor A assigned	PASS
TC-06	Create Enrollment with Instructor User = Instructor1	Owner changes to Instructor1; other instructors cannot see record	PASS
TC-07	Check Reports count matches records	Counts match records	PASS
TC-08	Dashboard refresh	Dashboard updates successfully	PASS

Screenshots Attached: Sharing Settings, Flow Active List, Object List, Validation Error, Owner Change Proof, Reports, and Dashboard.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data in a single Salesforce org
- Automated workflows reduce manual effort by approximately 80%
- Real-time reporting and dashboards
- Secure private sharing – instructors see only their enrollments
- Scalable and cloud-based 24/7

Disadvantages

- Salesforce dependency and licensing cost
- Requires initial configuration effort
- Needs training for non-technical staff

9. CONCLUSION

- SECMS successfully streamlines the entire student academic lifecycle from registration to course completion. By leveraging Salesforce automation—including Validation Rules, Flows, Approval Process, and OWD Private Sharing via Owner Change Flow—the system improves data accuracy, operational efficiency, and decision-making. The project demonstrates how Salesforce can transform manual academic processes into a

scalable, reliable, and user-friendly digital solution. All milestones have been implemented and tested.

10. FUTURE SCOPE

- Payment gateway integration (Razorpay, Stripe) for online fees
- Student self-service portal via Experience Cloud
- Analytics-driven insights with Einstein Analytics
- Mobile access via Salesforce Mobile App
- Certificate generation automation
- Integration with LMS

11. APPENDIX

Item	Details
Source Code	Salesforce Configuration (Flows, Objects, Validation Rules)
Dataset	Salesforce Custom Objects
GitHub	https://github.com/pdharshu265-dev
Demo Video	https://drive.google.com/file/d/1qAhAmp5QUOpQooPwYtHvH486TBdd8-Tb/view?usp=sharing
Developer Org	https://orgfarm-27651f0434-dev-ed.develop.lightning.force.com
Team Members	4 Members
Members Name	Bharathi. K Dharani Sri. G Dharshini. S Dhivya. C Dharshini. P
Project Type	Salesforce Administrator with Automation
Prepared By	Dharshini. P, Dr. M. G. R Government Arts and Science College for Women, Villupuram.
Date	September 2026

THANK YOU